Validation: the measurement pipeline passed everywhere; one designed control (S-conflict) failed

	Infrastructure I1–I9	Recovery checks (frozen thresholds; each must hold in all 3 orderings)						Descriptive flags
S-single	✓ 9/9	✓ B1	✓ B2	✓ B3	✓ B4	✓ B5		none
S-redundant	✓ 9/9	✓ BR1	✓ BR2	✓ BR3	✓ BR4	✓ BR5	✓ BR6	none
S-conflict	✓ 9/9	✗ **BC1**	✓ BC2	✗ **BC3**	✗ **BC4**	✗ **BC5**		φ(E1), φ(E2) vary by ordering; full-set argmax varies
Macro A	✓ 9/9	realistic scenario: descriptive only (no pass/fail recovery checks)						φ(E3) varies by ordering; full-set argmax varies
Macro B	✓ 9/9	realistic scenario: descriptive only (no pass/fail recovery checks)						φ(E1) range 0.101
Macro C	✓ 9/9	realistic scenario: descriptive only (no pass/fail recovery checks)						none

✓ pass ✗ fail. Recovery checks: B = single decisive, BR = redundant, BC = conflict (frozen thresholds). Infrastructure checks passed in every scenario.